2027 UF Guide to AI in Science

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Preface

AI is radically redefining the nature of scientific inquiry — for better and for worse.

In Fall 2026, faculty, graduate students, and postdocs across multiple departments at the University of Florida came together to understand this urgent development. In a seminar titled *Collective [Artificial] Intelligence in Science*, we documented and discussed how AI is changing each component of our research workflow. This includes project management, idea development, data analysis, writing, and dissemination. To push our understanding, we alternated between the perspective of an AI zealot and an AI critic – identifying key resources, consensus statements, and areas of disagreement along the way. The result is a *human-curated and written* **2027 UF Guide to AI in Science**.

The seminar began with an introduction to big-team science – its rise[1], competitive advantages[2], and relevance to the current moment. For instance, the paper describing OpenAI's GPT-1 had 4 authors; the paper describing GPT-3 had 31 authors; and the paper describing GPT-5 had 483 authors (Figure 1). Similarly, a prominent AI-led cybersecurity attack of the online community, Hugging Face, involved coordinated action between thousands of AI agents[3]. After reviewing these developments, the course itself became a large-scale collaboration, dedicated to understanding how AI is changing the nature of science.

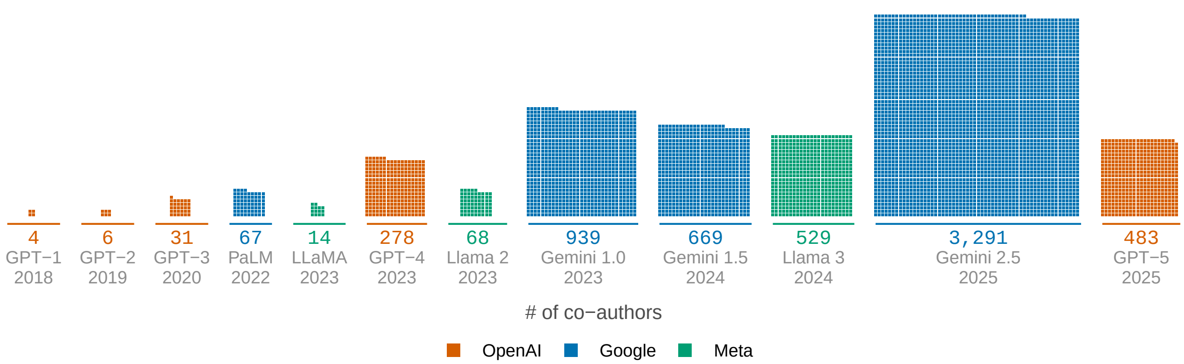

Figure 1: Credited co-authors on flagship AI model reports from OpenAI, Google, and Meta, 2018–2025. One square is one credited person; each block's area is its author list.

Each week followed a consistent structure (Figure 2). First, we would select a component of the research pipeline to examine (e.g., data analysis). Next, we would spend one week compiling resources for understanding the emerging role of AI in that part of the research pipeline. In class, we engaged in peer-to-peer teaching – all with the goal of identifying the top 3 resources to include in the guide. Last, we used a live onlined voting platform to solicit, debate, and vote upon key takeaways for the week (https://pol.is/). The result is the **2027 UF Guide to AI in Science**: a summary of our top resources, takeaways, and disagreements.

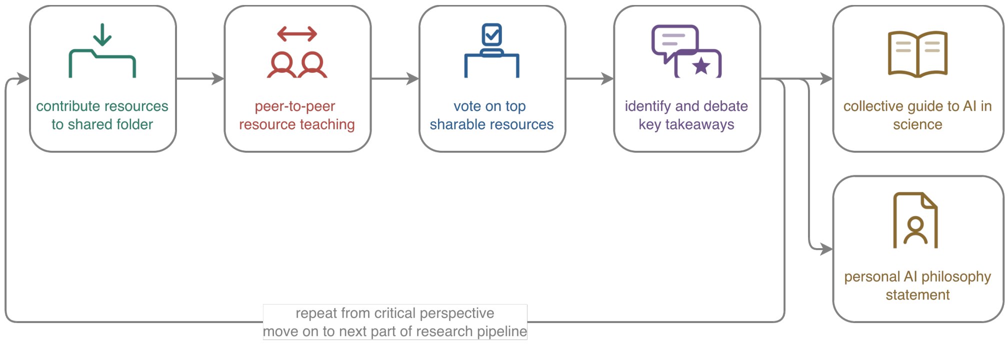

Figure 2: Structure of the UF Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science.

Ideation and Project Management

Ideation and Project Management: Using AI

Many researchers' first contact with AI tools is in the realm of ideation or project management (Figure 1). For example, one seminar participant noted that they have their research trainees submit weekly updates to a folder share with the LLM, Claude (Figure 2, panel A). This allows them to later prompt the model with questions about past conversations, challenges, decisions, and solutions (Figure 2, panel B). For example, the seminar participant demonstrated how Claude can be used to efficiently take stock of a trainee's progress, have a conversation with the LLM about potential future directions, and generate a simulated figure to send back to the trainee to convey next steps (Figure 2, panel C).

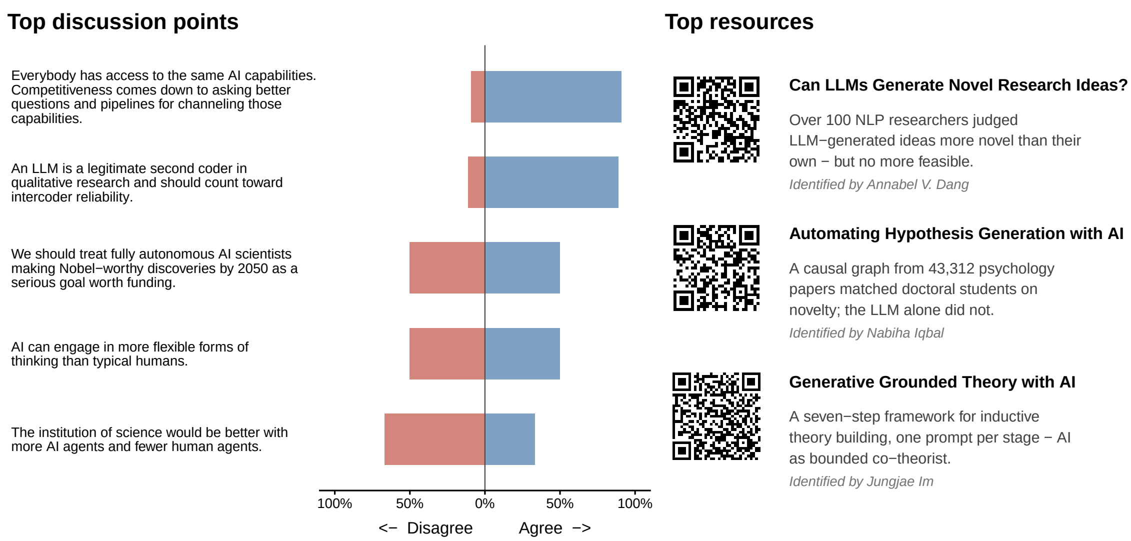

Figure 1: Overview of Week 2 of the Fall 2026 Collective [Artificial] Intelligence Seminar. Left: top discussion points, along with estimated level of agreement among seminar participants. Right: three most popular resources identified by seminar participants.

The resources compiled by seminar participants illustrate that AI is a tool that is increasingly being used to expand ideation and project management capabilities (Figure 1). For example, as early as 2025, researchers documented that LLMs were capable of generating research ideas that were judged more novel than human experts[4]. These capabilities can be further improved by integrating causal discovery algorithms. More specifically, LLMs can efficiently identify which causal relationships have been identified in past research. When mapped across studies, causal discovery algorithms can be used to identify gaps in causal network maps – i.e., potential future research directions[5]. Such results highlight the increasing sophistication in which researchers are deploying AI to organize information, extract ideas, and even develop theory[6].

Areas of consensus – and dissensus

Disagreement is common when researchers collaborate[7] – and our seminar discussion was marked by both consensus and dissensus.

In terms of consensus, there was broad agreement that LLMs can assist with ideation, project management, and the organization of information. There was also broad agreement that this is changing the nature of competition in science – placing more emphasis on who can better leverage these tools for ideation and execution.

Yet, there was clear disagreement about how far things should go. Seminar participants strongly disagreed about whether AI is reliably superior at ideation, whether the institution of science should shift to more AI-led work, and whether humanity should strive for fully autonomous AI scientists. These disagreements echo throughout the scientific landscape, with recent back-to-back Nature pieces on both the capabilities[8] and limitations[9] of science with fewer humans and more AI. These limitations became the focus of the following week of the Fall 2026 Collective [Artificial] Intelligence Seminar (see Section).

Figure 2: A seminar participant shared an LLM-assisted workflow to compile past meeting notes from their trainee (Panel A), review progress (Panel B), and mock a figure illustrating future directions (Panel C).

Ideation and Project Management: Critiquing AI

In February 2026, an engineer running an online educational program used Claude Code to help launch a new side project. Instead of launching something new, they ultimately deleted everything they had already built: servers, networks, and years worth of datasets. The failure point, the engineer would later suggest, was over-automation[10]. The engineer never checked if backups could be restored; they never pushed back on questionable recommendations from the coding agent; and they allowed it to run irreversible commands completely unsupervised. Issues like these are bound to become more common as scientists continuously rely more on AI — the focus of Week 3 of the Fall 2026 Collective [Artificial] Intelligence Seminar (Figure 1).

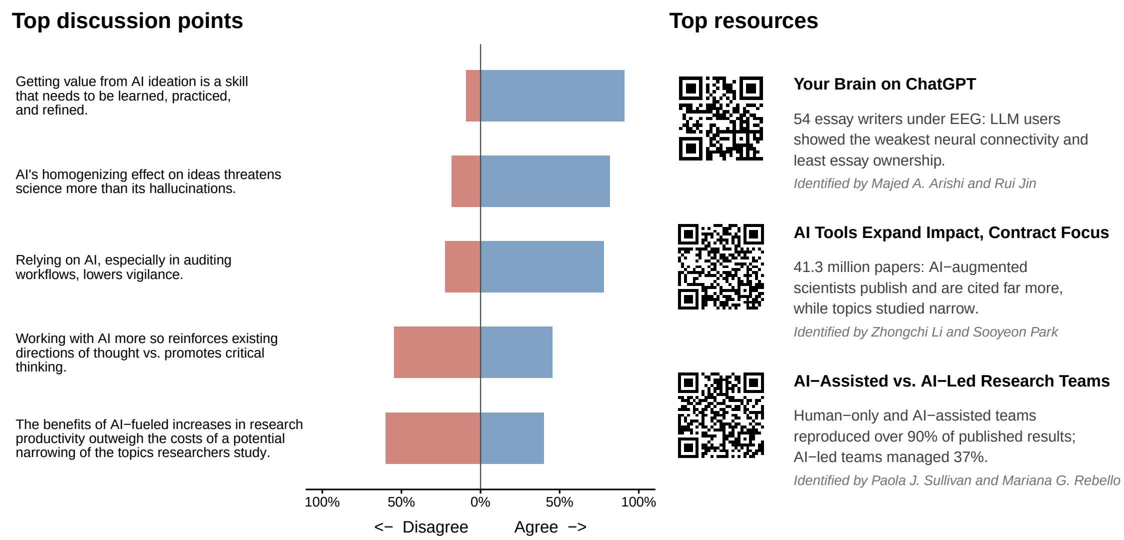

Figure 1: Overview of Week 3 of the Fall 2026 Collective [Artificial] Intelligence Seminar. Left: top discussion points, along with estimated level of agreement among seminar participants. Right: three most popular resources identified by seminar participants.

This week's most popular resources focused on the pitfalls of using AI for ideation and project management (Figure 1). For example, one study used electroencephalogram to measure brain connectivity patterns while writing alone, with a search engine, or with a LLM. Results suggested that LLM users exhibited the lowest levels of distributed brain activity — an index of lower cognitive engagement[11]. Similarly, when researchers rely more on AI for routine task like reproducibility checks, they catch fewer errors, propose weaker robustness checks, and ultimately move slower[12]. As scientists continue this shift towards AI, collective implications emerge. For instance, an analysis of 41 million research papers suggests that the adoption of AI tools and methods increases productivity, impact, and career outcomes for individuals — but at the expense of an overall narrowing of the scope of the science[13] (Figure 2).

Areas of consensus – and dissensus

Particular after focusing a prior week on the benefits of AI, seminar discussion was marked with areas of consensus and dissensus.

In general, there was strong agreement that concerns about AI have evolved beyond hallucination proneness[14]. Decreased vigilance and a collective narrowing of science received more attention among seminar participants — with near unanimous agreement that maximizing value from AI requires training, practice, and refinement. Yet, there was clear disagreement about the magnitude of the issue. Does AI hinder more than help with critical thinking? In the midst of widespread concerns about the robustness of existing research practices[15], to what extent should we aim for increases in productivity? Last, is a narrowing of the scope of science necessarily a bad tradeoff[16]? Depression, addiction, cancer, dementia, climate change, and polarization are arguably enduring and unsolved problems. Whether those problems are solved quicker through increases in individual productivity, decreases in collective scope, and a faster overall science remains an open question.

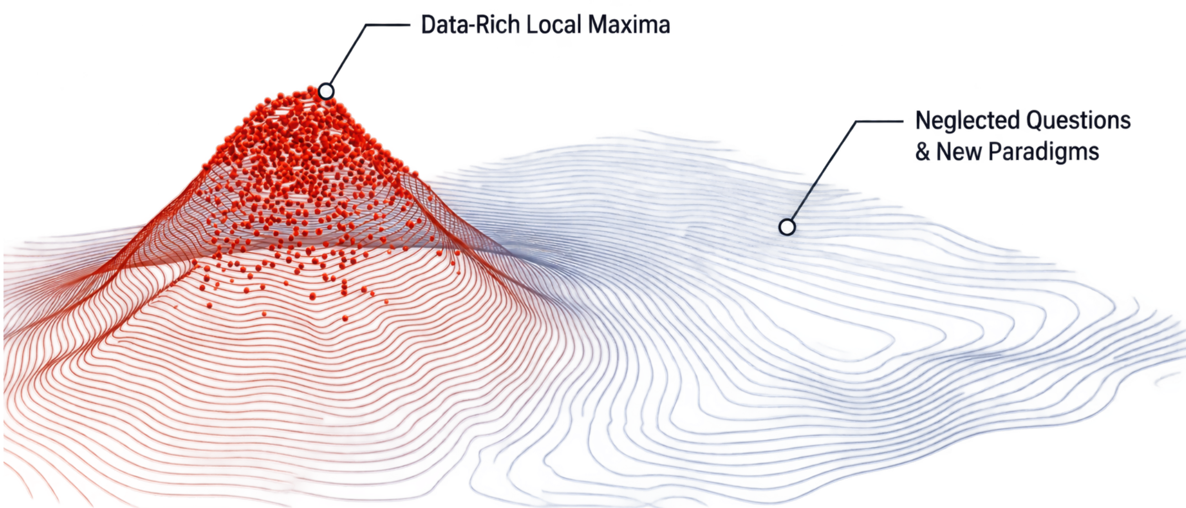

Figure 2: Illustration of the narrowing of the scope of science, created by Zhongchi Li and Sooyeon Park

Literature Review and Synthesis

Literature Review and Synthesis: Using AI

Covered in Week 4 of the seminar (September 10, 2026). Estimated date of arrival: September 20, 2026.

Literature Review and Synthesis: Critiquing AI

Covered in Week 5 of the seminar (September 17, 2026). Estimated date of arrival: September 27, 2026.

Data Collection and Generation

Data Collection and Generation: Using AI

Covered in Week 6 of the seminar (September 24, 2026). Estimated date of arrival: October 4, 2026.

Data Collection and Generation: Critiquing AI

Covered in Week 7 of the seminar (October 1, 2026). Estimated date of arrival: October 11, 2026.

Analysis and Modeling

Analysis and Modeling: Using AI

Covered in Week 8 of the seminar (October 8, 2026). Estimated date of arrival: October 18, 2026.

Analysis and Modeling: Critiquing AI

Covered in Week 9 of the seminar (October 15, 2026). Estimated date of arrival: October 25, 2026.

Writing and Communication

Writing and Communication: Using AI

Covered in Week 10 of the seminar (October 22, 2026). Estimated date of arrival: November 1, 2026.

Writing and Communication: Critiquing AI

Covered in Week 11 of the seminar (October 29, 2026). Estimated date of arrival: November 8, 2026.

Review, Outreach, and Dissemination

Review, Outreach, and Dissemination: Using AI

Covered in Week 12 of the seminar (November 5, 2026). Estimated date of arrival: November 15, 2026.

Review, Outreach, and Dissemination: Critiquing AI

Covered in Week 13 of the seminar (November 12, 2026). Estimated date of arrival: November 22, 2026.

Personal AI Philosophy Statements

Covered in Week 14 of the seminar (November 19, 2026). Estimated date of arrival: November 29, 2026.

References

1. Coles, N. A., Hamlin, J. K., Sullivan, L. L., Parker, T. H. & Altschul, D. Build up big-team science. *Nature* **601**, 505–507 (2022).
2. Coles, N. A., Goes Braga Takayanagi, J. F., Grant, G. L. & Basnight-Brown, D. M. The rise, impact, and imbalances of big-team psychology. *Psychological Science* **37**, 287–297 (2026).
3. OpenAI. The hugging face incident and the road ahead. https://openai.com/index/hugging-face-incident-and-the-road-ahead/ (2026).
4. Si, C., Yang, D. & Hashimoto, T. Can LLMs generate novel research ideas? A large-scale human study with 100+ NLP researchers. in *International conference on learning representations* (eds Yue, Y., Garg, A., Peng, N., Sha, F. & Yu, R.) vol. 2025 94003–94092 (2025).
5. Tong, S., Mao, K., Huang, Z., Zhao, Y. & Peng, K. Automating psychological hypothesis generation with AI: When large language models meet causal graph. *Humanities and Social Sciences Communications* **11**, 896 (2024).
6. Schmitt, B., Hao, S., Pham, M. T. & Hofstetter, R. Generative grounded theory (GGT): Inductive theory building in the age of generative AI. *Journal of Consumer Psychology* https://doi.org/10.1002/jcpy.70036 (2026) doi:10.1002/jcpy.70036.
7. Coles, N. A. *et al.* Team scientists should normalize disagreement. *Science* **384**, 1076–1077 (2024).
8. Yamada, Y. *et al.* Towards end-to-end automation of AI research. *Nature* **651**, 914–919 (2026).
9. Nature. Why AI cannot do good science without humans. *Nature* https://www.nature.com/articles/d41586-026-01551-3 (2026).
10. Grigorev, A. Claude Code wiped our production database with a Terraform command. (2026).
11. Kosmyna, N. *et al.* Your brain on ChatGPT: Accumulation of cognitive debt when using an AI assistant for essay writing task. *arXiv preprint arXiv:2506.08872* https://doi.org/10.48550/arXiv.2506.08872 (2025) doi:10.48550/arXiv.2506.08872.
12. Brodeur, A. *et al.* AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science. *Proceedings of the National Academy of Sciences* **123**, e2524747123 (2026).
13. Hao, Q., Xu, F., Li, Y. & Evans, J. Artificial intelligence tools expand scientists' impact but contract science's focus. *Nature* **649**, 1237–1243 (2026).

14. Topaz, M., Roguin, N., Gupta, P., Zhang, Z. & Peltonen, L.-M. Fabricated citations: An audit across $2 \cdot 5$ million biomedical papers. *The Lancet* **407**, 1779–1781 (2026).
15. Munafò, M. R. *et al.* A manifesto for reproducible science. *Nature Human Behaviour* **1**, 0021 (2017).
16. Kuhn, T. S. The essential tension: Tradition and innovation in scientific research. in *The essential tension: Selected studies in scientific tradition and change* 225–239 (University of Chicago Press, Chicago, 1977). doi:10.7208/9780226217239-010.